MNC Weather Report

Technical Guide

Daily weather summaries and time-series charts from GMMF weather stations

Version 1.1 · Generated July 21, 2026

Workflow id: **mnc_weather_report**

1. Overview

The **mnc_weather_report** workflow retrieves weather station observations from the EarthRanger subject group **ER2ER - From GMMF**, extracts seven meteorological variables from each observation's JSON detail field, computes daily aggregates per weather station, and renders an interactive time-series line chart for each variable.

The workflow delivers:

- 1 daily weather summary CSV — all seven variables aggregated per station and date
- 7 line charts (HTML + PNG) — one per meteorological variable, coloured by weather station

Output summary

Output	Type	Description
weather_summary_table.csv	CSV	Daily per-station summary: sum precipitation, mean temperature/wind/humidity/pressure, max gusts
precipitation_readings_over_time	HTML / PNG	Daily total precipitation (mm) per weather station
temperature_readings_over_time	HTML / PNG	Daily mean surface air temperature (°C) per weather station
wind_speed_readings_over_time	HTML / PNG	Daily mean wind speed per weather station
wind_gusts_readings_over_time	HTML / PNG	Daily maximum wind gusts per weather station
soil_temperature_readings_over_time	HTML / PNG	Daily mean soil temperature (°C) per weather station
relative_humidity_readings_over_time	HTML / PNG	Daily mean relative humidity per weather station
atmospheric_pressure_readings_over_time	HTML / PNG	Daily mean atmospheric pressure per weather station

2. Dependencies

2.1 Python packages

As of the migration to the **ecoscope-platform** task/compiler runtime, the workflow's requirements are:

Package	Version	Channel
ecoscope-platform	>=2.15.0, <2.16.0	ecoscope-workflows
ecoscope-workflows-ext-custom	0.1.0rc14.*	ecoscope-workflows-custom
pydeck	0.9.2	conda-forge

Note: ecoscope-workflows-core, ecoscope-workflows-ext-ecoscope, ecoscope-workflows-ext-ste, and ecoscope-workflows-ext-mnc are not dependencies of this workflow — every task it uses (get_subjectgroup_observations, extract_value_from_json_column, summarize_df, draw_line_chart, html_to_png, gather_dashboard, etc.) resolves from ecoscope-platform or ecoscope-workflows-ext-custom alone.

2.2 Connection

Connection	Task	Purpose
EarthRanger	set_er_connection	Fetch weather station observations from the subject group 'ER2ER - From GMMF'

Note: This workflow does not require Google Earth Engine or any Dropbox file downloads.

2.3 Subject group (fixed)

The subject group name is hardcoded in the spec as **'ER2ER - From GMMF'** and is not user-configurable. All observations are fetched with filter: clean, raise_on_empty: false, include_details: false, include_subjectsource_details: false.

2.4 Grouper

The workflow uses an **empty grouper list** (groupers: []). Data is not split by any grouping dimension. The grouper is passed through to the temporal index and dashboard only.

3. Observation Ingestion and Variable Extraction

Weather observations are fetched as subject group observations and then processed through a sequential extraction chain. Each step reads from the **extra__observation_details** JSON field and appends one new float column.

3.1 Observation retrieval

Parameter	Value
Task	get_subjectgroup_observations
subject_group_name	'ER2ER - From GMMF' (hardcoded)
filter	clean
raise_on_empty	false
include_details	false
include_subjectsource_details	false

3.2 Sequential variable extraction

Task: **extract_value_from_json_column** — applied seven times in a chain, each reading from **extra__observation_details** and writing a new float column. Each step receives the output of the previous step as its input DataFrame:

Step	field_name_options	output_column_name	output_type
1	precipitation	precipitation	float
2	surface_air_temperature	temperature	float
3	wind_speed	wind_speed	float
4	wind_gusts	wind_gusts	float
5	soil_temperature	soil_temperature	float
6	relative_humidity	relative_humidity	float
7	atmospheric_pressure	atmospheric_pressure	float

Note: field_name_options is a list, allowing the extractor to try multiple possible key names in the JSON. Only the first matching key found in each record is used.

3.3 Date extraction and column renaming

Step	Task	Detail
1	extract_column_as_type	Extract **fixtime** as output_type: **date** into a new column **date**. This strips the time component for daily grouping in the summary table.

Step	Task	Detail
2	map_columns	Rename **extra__subject__name** → **weather_station** (raise_if_not_found: true). This column identifies which physical weather station produced each observation.
3	add_temporal_index	Add temporal index using time_col: **fixtime**, groupers: [], cast_to_datetime: true, format: mixed.

4. Daily Weather Summary

Task: **summarize_df**. Observations are aggregated to daily resolution per weather station using the following aggregation rules (reset_index: true):

Variable	Aggregation	Rationale
precipitation	sum	Total daily rainfall accumulation
temperature	mean	Average surface air temperature for the day
wind_speed	mean	Average wind speed for the day
wind_gusts	max	Peak gust recorded during the day
soil_temperature	mean	Average soil temperature for the day
relative_humidity	mean	Average relative humidity for the day
atmospheric_pressure	mean	Average atmospheric pressure for the day

The summary DataFrame is grouped by **[weather_station, date]** producing one row per station per day. It is persisted to disk as **weather_summary_table.csv** and reused as the input for all seven line charts.

5. Line Charts

Seven line charts are generated from the daily weather summary, one per meteorological variable. All use **draw_line_chart** with identical layout parameters, differing only in the y-axis column and label.

5.1 Shared chart configuration

Parameter	Value
smoothing	null
x_column	date
category_column	weather_station
line_kwargs shape	linear
title	null (no title displayed)
showlegend	true
fontsize	20
fontcolor	#2c3e50
plot_bgcolor	#f5f5f5 (light grey background)
xaxis title	Date
xaxis tickformat	%Y-%m-%d
xaxis tickangle	-45°
xaxis rangemode	auto
yaxis tickformat	.1f (1 decimal place)
legend_title	Weather Station
hovermode	x unified

Note: The precipitation chart additionally sets showgrid: true, gridcolor: '#e0e0e0', gridwidth: 1 on the x-axis, making the grid lines visible for easier reading of daily totals.

5.2 Per-chart y-axis and output filename

Chart	y_column	y-axis title	Output filename
Precipitation	precipitation	Precipitation (mm)	precipitation_readings_over_time.html
Temperature	temperature	Average Temperature (°C)	temperature_readings_over_time.html
Wind speed	wind_speed	Average wind speed	wind_speed_readings_over_time.html
Wind gusts	wind_gusts	Max wind gusts	wind_gusts_readings_over_time.html

Chart	y_column	y-axis title	Output filename
Soil temperature	soil_temperature	Average Temperature (°C)	soil_temperature_readings_over_time.html
Relative humidity	relative_humidity	Average humidity	relative_humidity_readings_over_time.html
Atmospheric pressure	atmospheric_pressure	Average pressure	atmospheric_pressure_readings_over_time.html

6. PNG Conversion

All seven chart HTML files are converted to PNG in a single **html_to_png** call, with all paths passed as a list:

Parameter	Value
html_path	List of all 7 chart HTML paths
width	1 280 px
height	720 px
full_page	false
device_scale_factor	2.0 (effective resolution: 2 560 × 1 440)
wait_for_timeout	1 ms (static Plotly HTML — no tile maps)
timeout	0 (no overall page timeout)
max_concurrent_pages	10 (up to 10 charts rendered in parallel)

Note: Batching all seven charts into one html_to_png call with max_concurrent_pages: 10 is more efficient than seven separate calls. The near-zero wait_for_timeout is sufficient because all charts are static Plotly HTML with no external tile dependencies.

7. Output Files

All outputs are written to **ECOSCOPE_WORKFLOWS_RESULTS**.

File	Description
weather_summary_table.csv	Daily per-station summary table with all 7 aggregated weather variables
precipitation_readings_over_time.html / .png	Daily total precipitation (mm) line chart per weather station
temperature_readings_over_time.html / .png	Daily mean surface air temperature (°C) line chart
wind_speed_readings_over_time.html / .png	Daily mean wind speed line chart
wind_gusts_readings_over_time.html / .png	Daily maximum wind gusts line chart
soil_temperature_readings_over_time.html / .png	Daily mean soil temperature (°C) line chart
relative_humidity_readings_over_time.html / .png	Daily mean relative humidity line chart
atmospheric_pressure_readings_over_time.html / .png	Daily mean atmospheric pressure line chart

8. Workflow Execution Logic

8.1 Per-task skip conditions

This workflow does not use a global **task-instance-defaults** block. Every task from observation retrieval onwards carries its own explicit skipif block:

Condition	Behaviour
any_is_empty_df	Skip this task if any input DataFrame is empty
any_dependency_skipped	Skip this task if any upstream dependency was skipped

8.2 Linear pipeline

The entire workflow is a single linear chain with no branching, mapvalues fan-out, or parallel processing of data. The seven variable extraction steps are strictly sequential — each step receives the enriched DataFrame from the previous step. The seven chart generation steps are logically independent but all read from the same shared **daily_weather** summary DataFrame.

8.3 No maps or spatial outputs

This workflow produces no EcoMaps, scatter layers, or spatial visualisations — only tabular CSV data and time-series line charts. There are no draw_map, create_scatterplot_layer, or zoom_map_and_screenshot calls.

8.4 Dashboard

The workflow concludes with **gather_dashboard**, which packages workflow details, time range, groupers, and a **widgets** list containing seven plot widgets — one **create_plot_widget_single_view** per chart, each wired from its persisted HTML file:

Widget title	Source
Precipitation	precipitation_widget
Temperature	temperature_widget
Wind Speed	wind_speed_widget
Wind Gusts	wind_gusts_widget
Soil Temperature	soil_temp_widget
Relative Humidity	rel_humidity_widget
Atmospheric Pressure	pressure_widget

Note: Each widget-creation task carries **skipif: conditions: [never]**, so it always runs and appears on the dashboard even if its underlying chart data ended up empty for the run.

9. Software Versions

Package	Version pinned in spec.yaml
ecoscope-platform	>=2.15.0, <2.16.0
ecoscope-workflows-ext-custom	0.1.0rc14.*
pydeck	0.9.2